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Oligodendrogliomas: Reproducibility and Prognostic Value of Histologic Diagnosis and Grading

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Abstract. Prognostic value of histological grading of oligodendroglial tumors is controversial and interobserver reproducibility in grading of these tumors is unknown. Seven neuropathologists and 6 surgical pathologists experienced in brain tumor diagnosis assessed 124 oligodendroglial tumors operated at the Mayo Clinic (1960–1990). Among histologic parameters upon which current oligodendroglioma grading systems are based, only high cellularity, presence of mitoses, microcalcifications, endothelial hypertrophy, endothelial proliferation, and necrosis appeared to be reproducible. Reproducible histologic features, based on consensus ratings among neuropathologists (defined as >60%), were evaluated for the association with cause-specific survival by fitting Cox regression models. By univariate analysis, a significant association with survival was found for age, high cellularity, presence of mitoses, endothelial hypertrophy and proliferation and necrosis. On multivariable analysis with a stepwise variable selection method, only age and presence of endothelial proliferation were found to be independently associated with survival with a discriminatory index of the model of 0.68. Mitotic index was significantly associated with survival based on the grading from each separate neuropathologist, but it was not based on consensus, most likely because this was classified as indeterminate in 54% of cases. Alternatively, ''models fit'' considering the assessment of single neuropathologists, identified a model based on age and on mitotic index with similar discriminatory indices of 0.69–0.7. Our study found few factors independently associated with cause specific-survival among morphological parameters. These findings are consistent with the present WHO stratification of oligodendrogliomas into low- and high-grade variants.

Key Words: Grading; Histology; Interobserver reproducibility; Oligodendroglioma; Prognosis.

INTRODUCTION

Among gliomas, oligodendrogliomas remain an intriguing and mysterious tumor group. Although once considered rare, current literature indicates they may actually be the second most common glioma in adults after glioblastoma multiforme. In a recent retrospective review of a large group of all gliomas addressing diagnostic accuracy and interobserver concordance, oligodendrogliomas represented 25% of the total (1).

Recent advances in molecular genetics as well as the discovery of the remarkable sensitivity of oligodendrogliomas to a specific chemotherapy regimen have attracted much attention (2, 3). Combined loss of chromosomes 1 p and 19 q arms appears to be strongly associated with the oligodendroglial phenotype and also to be an independent predictor of survival both in high- and low-grade tumors (2). Furthermore, these molecular alterations have been associated with both a favorable chemotherapeutic response and prolonged overall survival in patients with anaplastic (grade III) oligodendrogliomas (3).

At present, histopathologic features remain the diagnostic ''gold standard'' in every day practice. It is upon these that critical clinical management decisions and prognostication are based. Seventy years after the first description of oligodendrogliomas by Bailey and Bucy, nuclear and overall cellular morphology remain the essential diagnostic criteria for a diagnosis of oligodendroglioma (4). According to the 2000 WHO classification, oligodendrogliomas are defined as tumors composed primarily of cells resembling oligodendrocytes and are divided into 2 groups, low grade (WHO II) and high grade or anaplastic (WHO III)(5). Significant interobserver variability, even among experienced neuropathologists, has cast doubts upon the reproducibility and predictive value of histologic diagnosis and grading as applied to oligodendrogliomas (6–8). To further evaluate the accuracy, reproducibility, and predictive value of histologic diagnosis and grading, we studied 124 oligodendrogliomas representing the full spectrum of histologic malignancy. We reviewed the various morphologic schemes that have been proposed for their classification and grading (6, 9–13), and identified the critical features used in each scheme (Table 1). The first aim of the study was to evaluate interobserver reproducibility in the identification of these features. Observers included almost equal numbers of general surgical pathologists and neuropathologists in each group. Upon obtaining data on interobserver concordance in each group, we evaluated statistically reproducible parameters for their prognostic significance based

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TABLE 1
Commonly Used Grading Systems and Critical Features for Categories

Criteria for Categories	Pleo-morphism	NC ratio	Ana-plasia	Cellularity	Mitoses	Endothelial hyperplasia	Endothelial proliferation	Necrosis	Astro-cytic cells
Grading System									
Ringertz 1950									
Oligodendroglioma	−		−	+/++	−/+	−/+	−/+	−	+
Malignant Oligodendroglioma	+		+	++/+++	++/+++	+/++	+/++	+/++	+
Oligodendroglial GBM	+		+	++/+++	+++	++/+++	++/+++	++/+++	+
NC ratio = Nuclear/Cytoplasmic ratio; Absent = −; Present = +; Mild = +; Moderate = ++; Marked = +++									
Smith 1983									
Grade A Oligodendroglioma	−	low		low		−	−	−	<50%
Grade B Oligodendroglioma	+	low/high		low/high		−	−	−	<50%
Grade C Oligodendroglioma	+	high		high		+	+	−	<50%
Grade D Oligodendroglioma	+	high		high		+	+	+	<50%
In Grade B, high NC ratio and/or moderate-high cellularity are present in addition to pleomorphism; in Grade C vascular changes are present; in Grade D, necrosis is also present.									
Kros 1988									
Oligodendroglioma	−	low		low		−	−	−	<50%
Malignant Oligodendroglioma	−/+	low/high		high		−/+	−/+	−	<50%
Oligodendroglial GBM	+	high		high		+	+	+	<50%
WHO 1993									
Oligodendroglioma	−		−/+	+/++	−/+	−/+	−/+	−	<50%
Anaplastic Oligodendroglioma	+		+	++/+++	++/+++	++/+++	++/+++	−/+	<50%

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TABLE 2
Histologic Features Assessed by Participating Pathologists

ATYPIA			
Hyperchromasia	0 = absent 1 = present	if present	1 = mild 2 = moderate 3 = marked
Pleomorphism	0 = absent 1 = present		
Coarse chromatin	0 = absent 1 = mild 2 = moderate 3 = marked		
NC RATIO	0 = normal 1 = low 2 = high		
ANAPLASIA* *cytologic malignancy	0 = absent 1 = present	if present	1 = occasional 2 = half 3 = most
CELLULARITY	low moderate high	for each: predominant	0 = absent 1 = present 1 = low 2 = moderate 3 = high
MITOSES	0 = absent 1 = present	if present # × 10 HPF	1 = 1 2 = 2–5 3 = ≥6
SUBPIAL INFILTRATION	0 = absent 1 = present 9 = unknown		
INVASION LEPTOMENINGES	0 = absent 1 = present 9 = unknown		
MICROCYSTS	0 = absent 1 = present		
CALCIFICATION	0 = absent 1 = present	if present	1 = mild 2 = moderate 3 = marked
ENDOTHELIAL			
Hypertrophy	0 = absent 1 = present	if present	1 = mild 2 = moderate 3 = marked
Proliferation	0 = absent 1 = present	if present	1 = mild 2 = moderate 3 = marked
INCREASED VASCULARITY	0 = absent 1 = present		
NECROSIS	0 = absent 1 = present	if present: amount palisading macrophages	 1 = patchy 2 = extensive 0 = absent 1 = present 0 = absent 1 = present
Mini-gemistocytes Gemistocytes Fibrillary astrocytes	for each:	0 = absent 1 = ≤10% 2 = 11–50% 3 = >50%	
COMMENTS	0 = none 1 = yes	if yes, specify	

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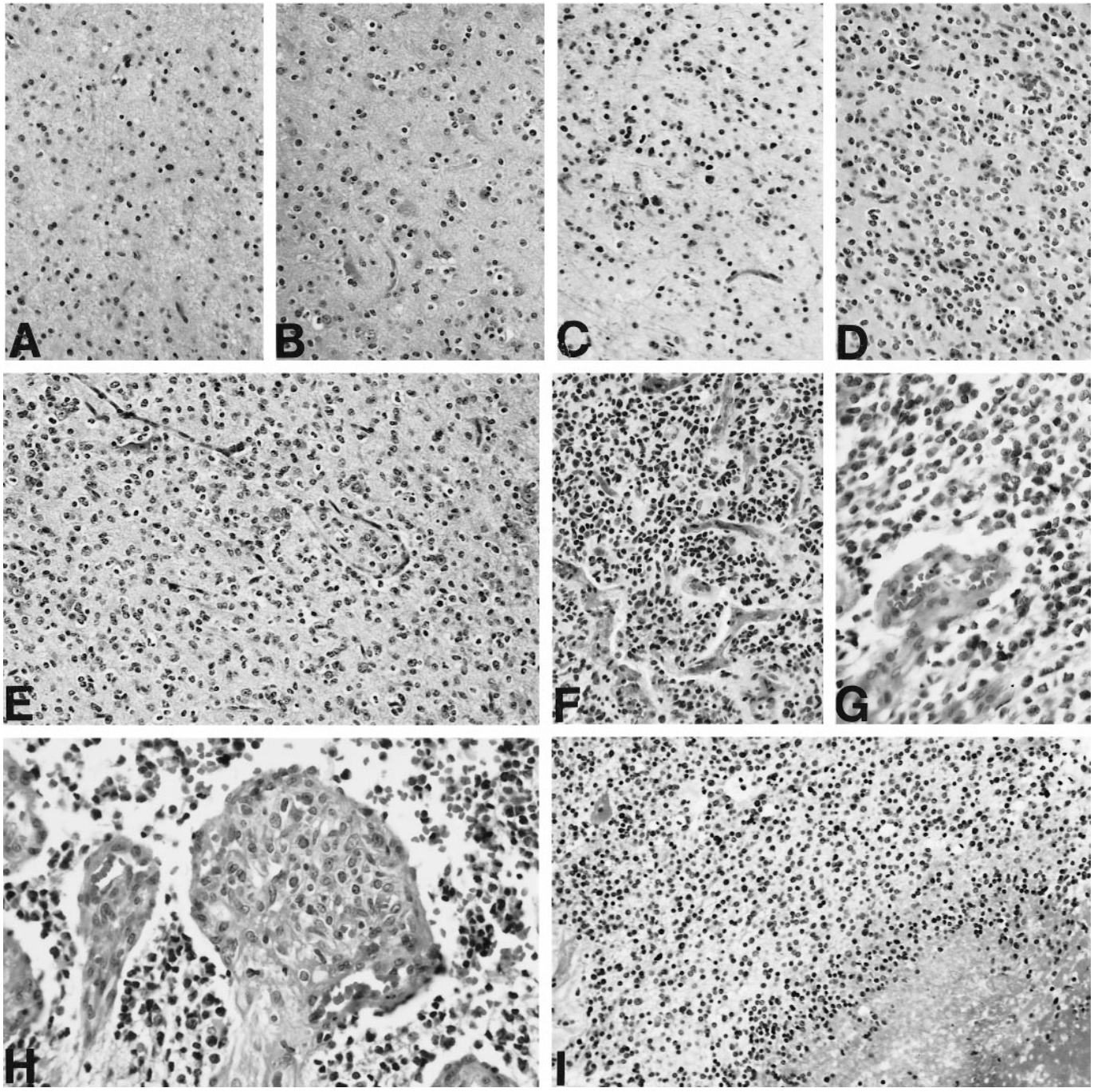


Fig. 1. The first 2 panels are examples of oligodendrogliomas without perinuclear halo artifact (A) and with perinuclear halo artifact (B). Note roundness and regularity of nuclei (B). Panel (C) shows scattered hyperchromatic cells. In some instances, they appear to ''pop-up'' out of the background. Showing the effect of prompt fixation, panel (D) features lack of the familiar perinuclear halo artifact associated with delayed fixation (E). Panel (E) illustrates the ''chickenwire'' vasculature of typical low-grade oligodendrogliomas that consists of thin vessel walls and inconspicuous endothelium. This is not endothelial hypertrophy. Panels (F) and (G) show examples of endothelial hypertrophy. The conspicuous vessel walls are thickened and pink. Although the endothelial cells are plump and impinge upon the lumen, they are not multilayered appearing. In panel (H), endothelial proliferation consists of apparent multilayering of endothelial cells. In this example the endothelium appears to be at least 2 layers thick. G: Necrosis accompanied by a rather modest increase in nuclei surrounding the focus i.e. ''palisading.''

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TABLE 3
Summary of Results: Prevalence and Intraclass Correlation Coefficient (ICC)

	Assessment of Inter-Rater Reliability Within Each Group of Pathologists			
	Neuropathologists		Surgical Pathologists	
Feature	Prevalence (%)*	ICC†	Prevalence (%)	ICC
Hyperchromasia	97	0.08	75	0.17
Pleomorphism	63	0.11	53	0.21
NC ratio	—	0.06	—	0.42
Anaplasia	41	0.34	57	0.15
Degree of anaplasia	—	0.45	—	0.30
Low cellularity	65	0.49	61	0.44
Moderate cellularity	75	0.38	77	0.28
High cellularity	39	0.65	46	0.57
Predominant cellularity	—	0.66	—	0.59
Mitoses	68	0.49	59	0.51
# of mitoses (× 10 HPF)	—	0.66	—	0.69
Microcysts	52	0.41	38	0.31
Microcalcifications	34	0.80	38	0.69
Endothelial hypertrophy	31	0.76	34	0.42
Degree of endothelial hypertrophy	—	0.75	—	0.52
Endothelial proliferation	20	0.52	19	0.55
Degree of endothelial proliferation	—	0.53	—	0.60
Hypervascularity	55	0.37	57	0.17
Necrosis	15	0.71	13	0.62
Degree of necrosis	—	0.75	—	0.68
Palisading necrosis	—	0.74	—	0.68
Macrophages	—	0.61	—	0.54
Minigemistocytes	15	0.30	7	0.18
Gemistocytes	7	0.13	7	0.18
Fibrillary astrocytes	42	0.11	38	0.05

* The prevalence of each feature assessed as either present or absent was calculated as the average of the prevalences as determined by each pathologist within the group.
† ICC, intraclass correlation coefficient.

on neuropathologist consensus. Additionally, we sought to determine which of the presently used grading schemes is maximally effective in prognostically stratifying oligodendroglial tumors and to formulate a working scheme based upon the data derived from the study.

MATERIALS AND METHODS

Patient Population and Characteristics

The study included neoplasms originally diagnosed as oligodendroglial neoplasms from 124 patients operated at Mayo Clinic from 1960 through 1990. The assessment of reproducibility was conducted on all 124 cases. However, survival analysis was limited to 122 cases because in 2 cases, slides were only available from the recurrence and not from the initial biopsy-resection. As reclassified recently according to the WHO classification (6), by 1 participating neuropathologist (BWS), the cases included 102 (82%) pure oligodendroglial tumors (66 WHO grade II; 36 grade III) and 22 (18%) mixed oligoastrocytomas (15 WHO grade II; 7 grade III). Clinical data included patient age and gender, extent of surgery, site and extent of tumor (1 lobe vs greater than 1 lobe), use of adjuvant radiation and/or chemotherapy, date of last follow-up or death, and cause of death. These data were obtained by chart review and by letters and telephone calls to patients, families, and physicians.

Assessment of Reproducibility in Histologic Diagnosis

Six surgical pathologists and 7 neuropathologists, including pathologists from different North American as well as European institutions, undertook histologic study of the entire study set. The same hematoxylin and eosin (H&E) slide set was circulated among the participating pathologists, all experienced in brain tumor diagnosis. Observers were asked to assess the cases according to the form reproduced in Table 2, which groups related histologic features from the various schemes presently used in classifying and grading oligodendrogliomas (6, 9–13). Written explanations, as well as kodachrome examples of some of the histologic features being assessed (Fig. 1), accompanied the H&E slide set and the study forms. Particular attention was given to the definition of vascular changes. Of these, 3 different types were considered. Increase in vascularity was defined as the presence of delicate pipe stem vasculature and rather thin vessel walls with inconspicuous endothelium (Fig. 1E). Endothelial hypertrophy was characterized by plump vessel walls, conspicuous due to their eosinophilia. The endothelial cells are not really piled up, but appear plump and impinge upon the lumen (Fig. 1F, G). Endothelial proliferation was defined as apparent multilayering (2–5 layers) of endothelial cells, not simply as increased vascularity or the aggregation of vessels with multiple thin-walled lumens (‘‘glomeruloid vessels’’) or

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TABLE 4
Comparisons of Consensus Ratings Between the 2 Groups of Pathologists for the Histologic Features With At Least Moderate Reproducibility

	Category	Neuropathologists consensus rating (%)	Surgical pathologists' consensus rating (%)	Bowker's test (2-sided p value)
High cellularity	Absent	56	52	NS*
	Present	35	40	
	Indeterminate	9	8	
Predominant cellularity	Low	23	15	0.004
	Moderate	43	40	
	High	19	26	
	Indeterminate	15	19	
Mitoses	Absent	23	38	<0.001
	Present	64	52	
	Indeterminate	14	10	
# of mitoses (× 10 HPF)	None	23	38	<0.001
	1	8	7	
	2–5	6	15	
	≥6	6	12	
	Indeterminate	56	28	
Microcalcifications	Absent	66	62	NS
	Present	32	33	
	Indeterminate	2	5	
Endothelial hypertrophy	Absent	68	68	NS
	Present	27	27	
	Indeterminate	5	5	
Endothelial proliferation	Absent	76	82	0.04
	Present	16	15	
	Indeterminate	8	3	
Necrosis	Absent	84	87	NS
	Present	13	11	
	Indeterminate	3	2	

* NS, not statistically significant, $p > 0.05$.

telangiectatic vessels (Fig. 1H). Each pathologist was asked to carefully read the instructions and review the illustrations prior to reviewing the histological sections. At the end of the review of each case, although the pathologist was not asked to provide a specific diagnosis, he/she had the opportunity of rejecting a tumor as ''non-oligodendroglial'' through a comment field. All cases were accepted.

Statistical analyses included calculation of a) the overall prevalence of each feature, b) chance corrected measures of agreement for each feature, estimated as intraclass correlation coefficient (ICC), and c) the consensus rating of each feature, separately for the 2 groups of pathologists.

The overall prevalence was determined for features assessed as either absent or present and calculated as the average of the prevalences as determined by each pathologist within each group (surgical pathologists and neuropathologists). Since we were dealing with both simple binary responses (absent/present) and ordinal categories (mild, moderate, marked; low, moderate, high, etc) as well as with multiple observers (> 2), determination of ICC was considered the most appropriate method of estimating chance corrected measures of agreement. The latter was calculated based on the variance components obtained from fitting two-way random effects analysis of variance (ANOVA) models (14). This approach is effectively equivalent to the weighted kappa statistic, thereby taking into account prevalence of findings as well as giving partial credit to varying degrees of disagreement. ICC values can be interpreted using the same guidelines used for weighted kappa statistics: ICC values < 0 correspond to agreement less than due to chance, ICC ≤ 0.4 signify slight or fair agreement, ICC within 0.41–0.6 indicate moderate agreement, ICC within 0.61–0.8 indicate substantial agreement, and ICC > 0.8 indicate almost perfect agreement. Parameters in which at least moderate reproducibility (ICC > 0.4) was obtained within both groups of pathologists (surgical pathologists and neuropathologists) were further analyzed by comparing the consensus rating between the 2 groups using Bowker's test for symmetry (2-sided p value). Bowker's test for symmetry is an extension of McNemar's test for comparing correlated proportions when there are more than 2 unordered response categories (15). Consensus was arbitrarily defined as a rating endorsed by at least 60% of the pathologists within a group (4 of 6 among surgical pathologists and 5 of 7 among neuropathologists). The consensus rating for a feature was classified

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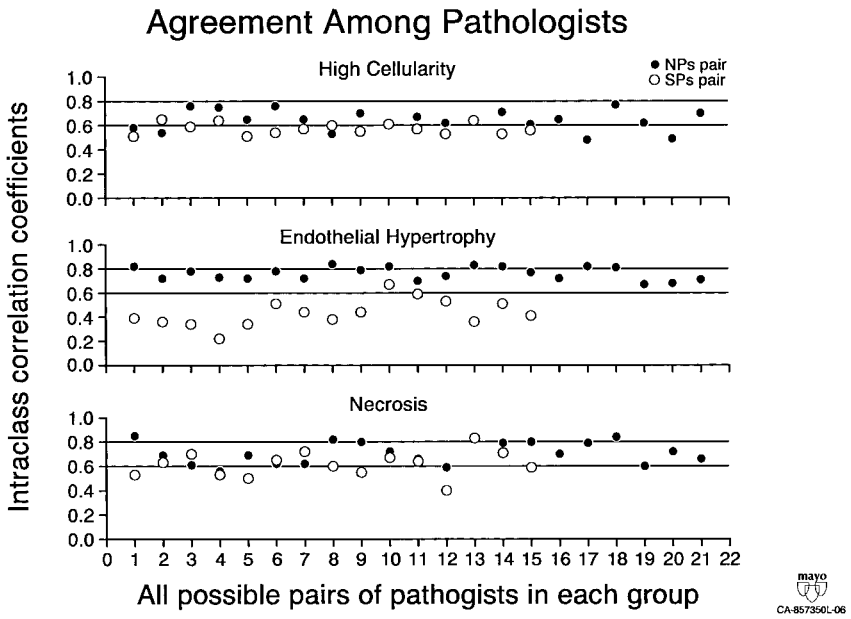


Fig. 2. The graphs illustrate ICC for all possible pairs of pathologists in both groups (black circles indicate neuropathologist's [NP] pairs, open circles indicate surgical pathologist's [SP] pairs). Data regarding presence of high cellularity (top), endothelial hypertrophy (middle), and necrosis (bottom) are provided. ICC between 0.61 and 0.8 are considered to reflect ''substantial agreement.'' NPs appear to agree with each other slightly more often than SPs based on their higher ICC.

TABLE 5
Summary of Univariate Associations with Cause-Specific Survival

Clinical feature	Univariate analysis (log rank test)	Histologic feature*	Univariate analysis (log rank test)
Age	0.0001	High cellularity	0.001
Gender	NS†	Predominant cellularity	NS
Extent of surgery	0.038	Mitoses	0.002
Radiation Therapy	0.051	# of mitoses	NS
Chemotherapy	NS	Endothelial hypertrophy	0.0001
Treatment (None, RT, Chemo, Chemo + RT)	NS	Endothelial proliferation	0.0001
Site (frontal, temporal, parietal)	NS	Microcalcifications	NS
One lobe vs >1 lobe	NS	Necrosis	0.01

* The histologic features are based on the consensus ratings by the neuropathologists.
† NS, not statistically significant, $p > 0.05$.

as indeterminate if a rating was not endorsed by at least 60% of the observers.

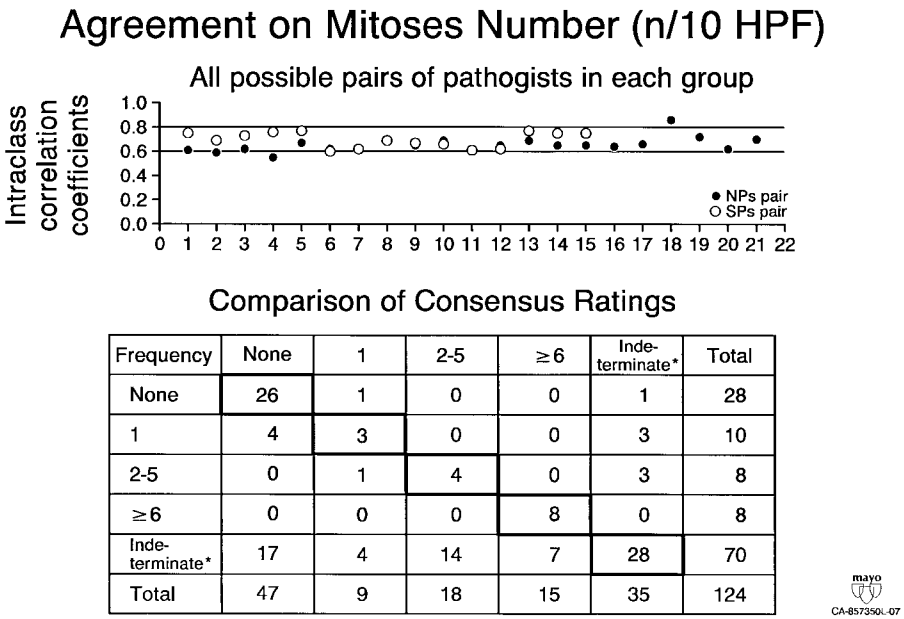
Survival Analysis and Evaluation of Predictive Value

Survival was calculated from date of diagnosis to date of death or last follow-up. Overall survival (death due to any cause) and cause-specific survival were estimated using Kaplan-Meier method. The Cox proportional hazards model was used to assess the strength of association between clinicopathological parameters and cause-specific survival. Among histologic parameters, only parameters with at least moderate reproducibility (ICC >0.4) were assessed using the consensus ratings of the neuropathologists. Each of the clinicopathological parameters was first evaluated univariately, then a multivariable Cox regression model using a stepwise variable selection method was fit to identify a set of features independently associated with cause-specific survival. All calculated p values are 2-sided and p values <0.05 were considered statistically significant. The discriminatory ability of the various models were summarized by calculating a discriminatory (or concordance) index described by Harrell et al for time to event models with censoring (16). The index ranges from 0 to 1; a value of 0.5 indicates predictive ability no different than due to chance alone, while a value of 1 indicates perfect predictive ability.

Effectiveness/Modeling

Strictly adhering to published grading criteria, we attempted to classify each case by the various schemes shown in

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Frequency	None	1	2-5	≥6	Inde-terminate*	Total
None	26	1	0	0	1	28
1	4	3	0	0	3	10
2-5	0	1	4	0	3	8
≥6	0	0	0	8	0	8
Inde-terminate*	17	4	14	7	28	70
Total	47	9	18	15	35	124

Fig. 3. The upper graph illustrates ICC for all possible pairs of pathologists in both groups (NPs and SPs) (black circles indicate NP's pairs, open circles indicate SP's pairs) regarding number of mitoses per 10 HPF. The lower graph illustrates comparisons of consensus ratings for a number of mitoses. Asterisk indicates indeterminate consensus in each group. Cumulative data for NPs is reported on the right side. Cumulative data for surgical pathologists is reported at the bottom. Neuropathologists are less likely not to find mitotic figures than surgical pathologists (28 of 124 NPs vs 47 of 124 SPs).

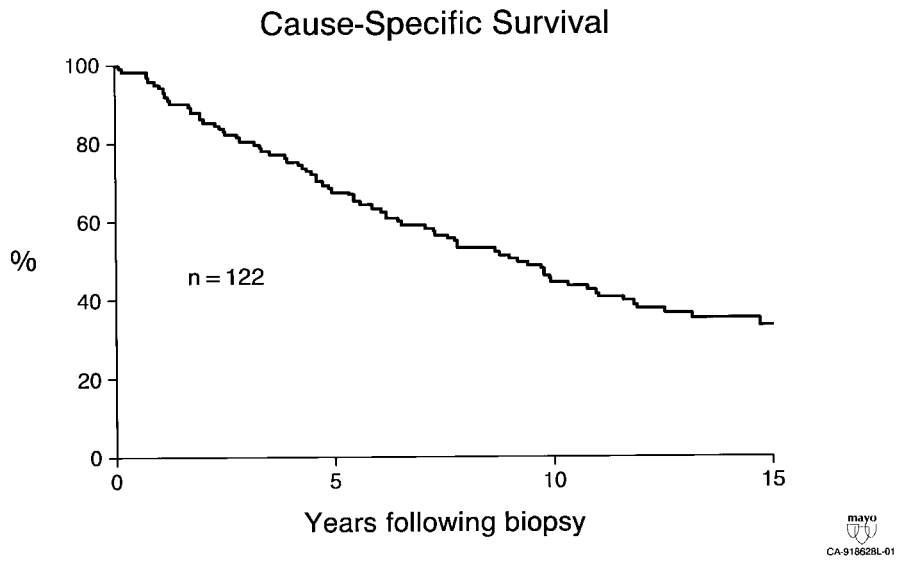


Fig. 4. The cause-specific survival at 5 and 10 years based on the 122 patients was 67% and 42%, respectively. The number of patients still at risk at 5, 10, and 15 years was 75, 46, and 17, respectively.

Table 1. ''Classification'' of each case was done automatically by computer based on the descriptive data provided by each pathologist.

RESULTS

Assessment of Reproducibility

Tables 3 and 4 summarize the prevalence and ICC for each feature, as well as consensus ratings, as determined separately for each group of pathologists, surgical and neuropathologists. Reproducibility appeared to be moderate or substantial for only some of the features upon which current oligodendroglioma grading schemes are based. These features include the presence of high cellularity, the predominant cellularity, the presence of mitoses and the number of mitoses per 10 HPF, the presence of microcalcifications, endothelial hypertrophy,

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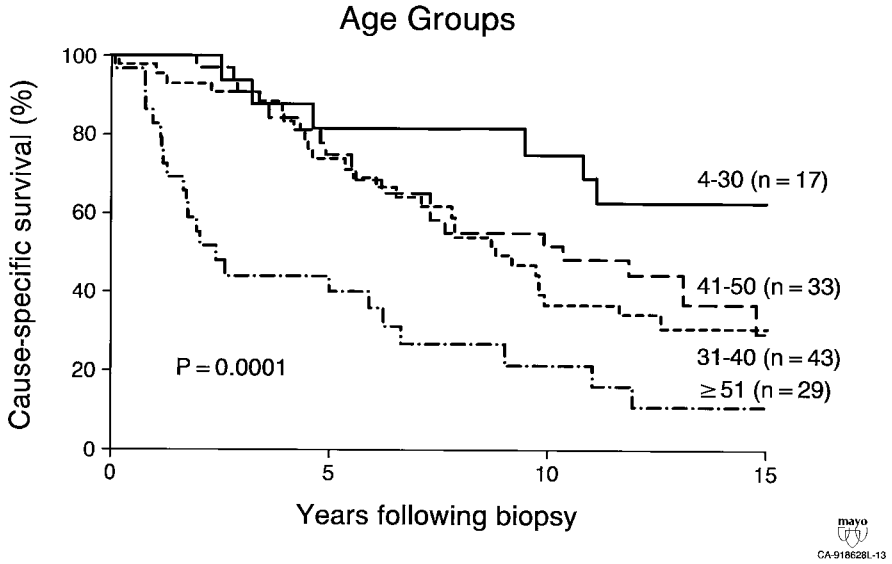


Fig. 5. The cause-specific survival at 5 years, stratified by the 4 age groups, was 81%, 74%, 75%, and 40%, respectively. Age, evaluated as a continuously scaled variable, was significantly associated with cause-specific survival ($p < 0.001$).

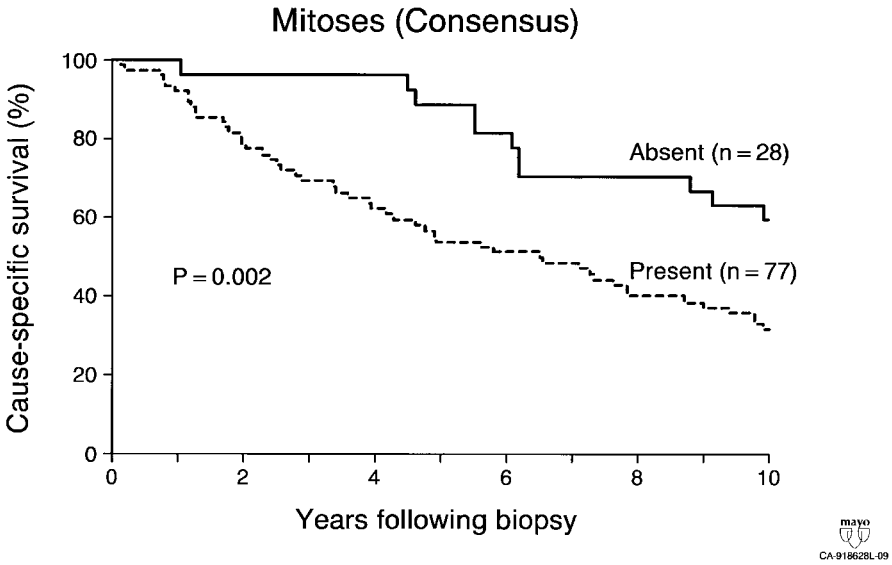


Fig. 6. The cause-specific survival at 5 years, stratified by the presence of mitoses as defined by consensus among the neuropathologists, was 89% (absent), 54% (present), and 87% (indeterminate, n = 17). The p value was based on comparing all 3 consensus categories although the survival curve for the indeterminate cases is not shown.

endothelial proliferation and necrosis. Agreement among surgical and neuropathologists, separately, is further illustrated in Figures 2 and 3, which show the ICC for all possible pairs of pathologists in both groups. In this study, neuropathologists seem to agree with each other slightly more often than surgical pathologists. The results of comparing consensus ratings for number of mitoses per 10 HPF are illustrated in the lower part of Figure 3. Although in our study neuropathologists were more likely to find a tumor with mitotic activity than surgical pathologists when assessing mitotic indices, the number of cases in which consensus was indeterminate was very high (Fig. 3).

Survival Analysis

Survival analysis was limited to 122 cases, since in 2 cases (of the original 124) slides from the initial biopsy-resection were not available. The study group included 48 women and 74 men in which initial biopsy-resection

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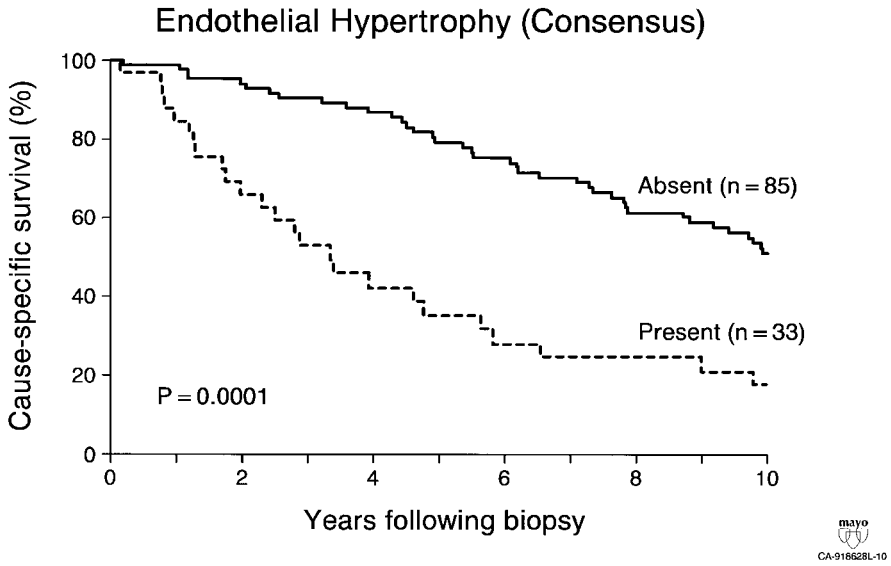


Fig. 7. The cause-specific survival at 5 years, stratified by the presence of endothelial hypertrophy as defined by consensus among the neuropathologists, was 79% (absent), 35% (present), and 50% (indeterminate, n = 4). The p value was based on comparing all 3 consensus categories although the survival curve for the indeterminate cases is not shown.

slides were available. The mean age at time of first diagnosis was 40 ± 14 (mean ± SD), median 41 years. The length of the follow-up was of 8.7 ± 6.8 (mean ± SD) years, 7.5 years (median). At time of last follow-up 94 patients had died, 80 as a result of the disease. The length of follow-up of the 28 patients known to be alive at termination of data collection was 15.5 ± 5.7 years (mean ± SD) and 13 years (median). In all but 2 cases, the most recent follow-up data were obtained within the last year. Only 1 surviving patient had a short follow-up (3.4 years), the remainder having follow-ups exceeding 10 years.

Overall survival at 5 and 10 years was 62% and 38% respectively. Cause-specific survival at 5 and 10 years was 67%, and 42%, respectively, as illustrated in Figure 4. The median cause-specific survival was 8.8 years. Results of the univariate analysis of factors for their association with cause-specific survival are summarized in Table 5. Among clinical features, age was strongly associated with cause-specific survival ($p < 0.001$); the 10-year cause-specific survival estimates for patients diagnosed under the age of 30, 30–50, or over the age of 50 years were 75%, 43%, and 21%, respectively, as illustrated in Figure 5. Cause-specific survival was also significantly associated with the extent of surgery ($p = 0.038$); patients selected to have a gross total resection or a biopsy tended to have better survival than those with a subtotal resection (10-year cause specific survival 57%, 48%, and 33%, respectively). A detailed analysis of clinical data and its correlation with histology and survival will be the subject of a separate publication.

Among the histologic features examined, univariate analyses based on consensus data among neuropathologists found high cellularity and the presence of mitoses, endothelial hypertrophy, endothelial proliferation, and necrosis to be significantly associated with cause-specific survival. Vascular changes, including endothelial hypertrophy and endothelial proliferation had the strongest association univariately with survival ($p < 0.001$). The cause-specific survival stratified by presence of high cellularity, mitoses, endothelial hypertrophy, endothelial proliferation, and necrosis are illustrated in Figures 6–10.

The absence of a statistically significant association between the number of mitoses per 10 HPF and patient survival appears to be paradoxical given the significant association between presence of mitotic activity and survival (Table 5). This can be explained and attributed to our strict definition of consensus (greater than 60%, i.e. at least 5 of 7 neuropathologists). Minimal disagreement among neuropathologists, usually absence of mitoses vs presence of a single mitosis, caused a large number of cases to be classified by consensus as indeterminate. Consensus among neuropathologists, according to our definition, could be reached in only 56 cases, including 28 cases without mitoses, 10 cases with only 1 mitosis per 10 HPF, 10 with 2–5 mitoses, and 8 with ≥ 6. Whereas absence of an association based on the consensus data was due mainly to the classification of numerous cases in the indeterminate category, a statistically significant association between mitotic index and cause-specific survival was found when results were examined according to individual

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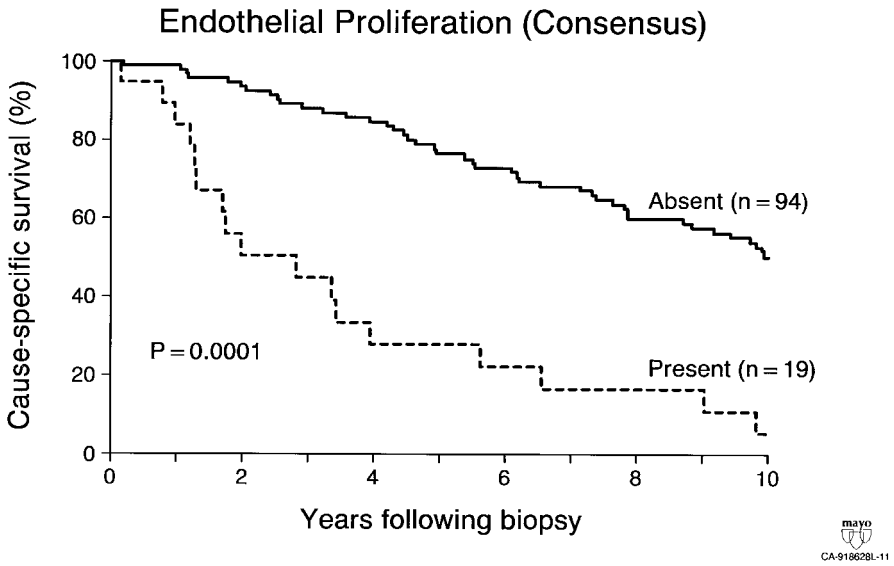


Fig. 8. The cause-specific survival at 5 years, stratified by the presence of endothelial proliferation as defined by consensus among the neuropathologists, was 77% (absent), 28% (present), and 44% (indeterminate, n = 9). The p value was based on comparing all 3 consensus categories although the survival curve for the indeterminate cases is not shown.

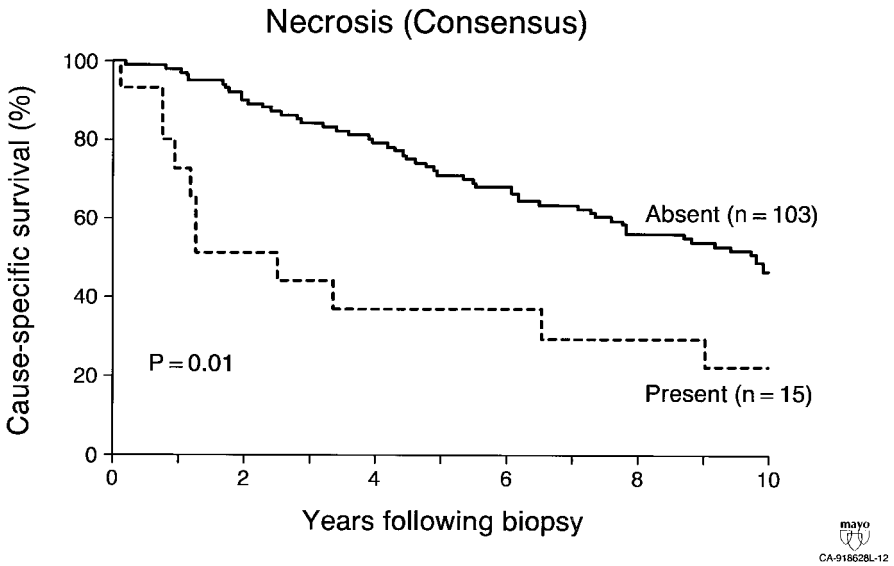


Fig. 9. The cause-specific survival at 5 years, stratified by the presence of necrosis as defined by consensus among the neuropathologists, was 71% (absent), 36% (present), and 67% (indeterminate, n = 4). The p value was based on comparing all 3 consensus categories although the survival curve for the indeterminate cases is not shown.

neuropathologists. There was a statistically significant association for 6 of the 7 individual neuropathologists (Table 6). Figure 10 illustrates the cause-specific survival curves for Neuropathologist 1's (NP1) assessment of number of mitoses per 10 HPF. Cause-specific survival curves for NP2 and NP3 were similar to those illustrated for NP1 (data not shown).

Based on a multivariable model using a stepwise variable selection method considering the clinical features and histologic features defined by consensus among neuropathologists, only age and endothelial proliferation were identified as independently associated with cause-specific survival. A 10-year increase in age was associated with a hazard ratio of 1.5 (95% confidence interval 1.2–1.8, $p < 0.001$). Presence of endothelial proliferation was associated with a hazard ratio of 2.7 (95% confidence interval 1.5–4.8, $p < 0.001$). The discriminatory index of this model was 0.68.

Additional models emerged when multivariable modeling was repeated considering the histology assessments by single neuropathologists. For example, in the case of NP1 and NP2, a model including age and number of

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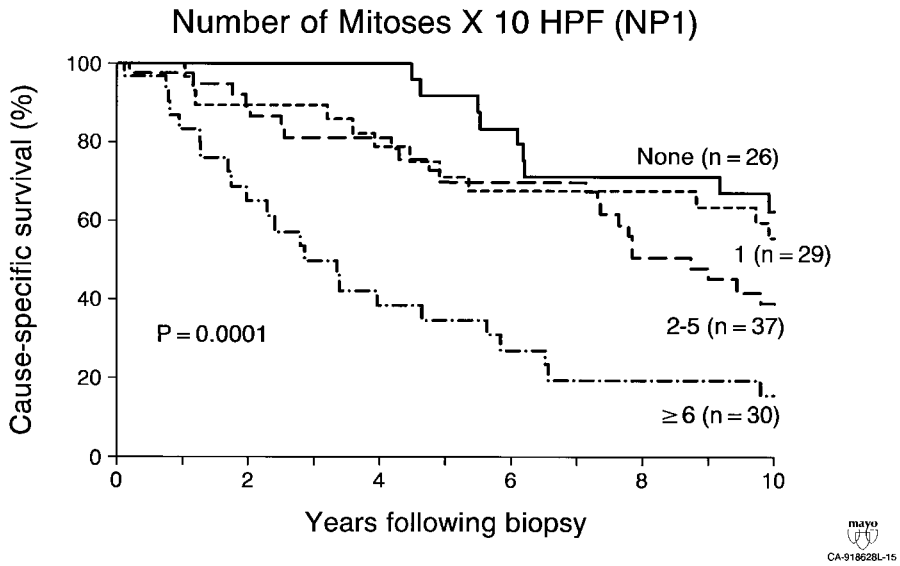


Fig. 10. The cause-specific survival at 5 years, stratified by the presence of the number of mitoses × 10 HPF for a single neuropathologist (NP1 as an example), was 92% (none), 71% (1), 70% (2–5) and 24% (≥6).

TABLE 6
Correlation Between Number of Mitoses and Survival of Neuropathologists 1 to 7

Histologic feature	P value from univariate analysis of the ratings from each of the 7 neuropathologists							
	Consensus	1	2	3	4	5	6	7
Mitoses	<0.0001	0.002	0.007	0.006	0.003	0.02	0.001	<0.001
Number of mitoses	NS*	<0.001	<0.001	<0.001	0.002	NS	<0.001	0.002

* NS, not statistically significant, $p > 0.05$.

mitoses appeared to be similar to a model including age and endothelial proliferation. Respective discriminatory indices for these 2 models were 0.69 and 0.67 for NP1, and 0.69 and 0.66 for NP2. For NP3 there were 2 possible models: 1 including age and number of mitoses, and 1 including age and endothelial hypertrophy, with respective discriminatory indices of 0.7 and 0.68.

Evaluation of Grading Schemes

Based on the descriptive data provided by each pathologist, the percentage of cases that could not be classified strictly according to published grading criteria was high (Table 7). The presence of poorly reproducible parameters among published grading criteria contributes to this difficulty. As shown in Table 7, after pleomorphism and anaplasia were no longer included as grading criteria, there was a large increase in the percentage of tumors classifiable according to the 1993 WHO classification scheme.

DISCUSSION

Uniform and reproducible classification and grading of gliomas is essential to providing patients with information regarding the natural history and prognosis of their disease as well as a rational basis upon which new, aggressive treatment regimens can be selected and their efficacy tested. Although there has been considerable debate on the subject of glioma classification and grading, particularly as it relates to oligodendrogliomas, morphological assessment remains the ''gold standard'' in every day practice (8). In the present study, we examined the ability of a group of surgical and neuropathologists to reproducibly identify histological features upon which systems of classification and grading of oligodendrogliomas are based. It is apparent from our results that numerous parameters listed among diagnostic and grading criteria of this tumor type (e.g. cellular pleomorphism, nuclear hyperchromasia, coarse chromatin, high NC ratio, and anaplasia) are, perhaps due to vagueness of definition, not easily reproduced. On the other hand, a more

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TABLE 7
Classification of Cases according to Published Garding Schemes

	Neuropathologists*							Surgical Pathologists					
	1	2	3	4	5	6	7	1	2	3	4	5	6
Grading system													
Ringertz 1950													
Indeterminate	87	60	55	72	74	59		88	90	61	62	40	60
Oligodendroglioma	1	21	26	2	6	34		8	2	20	24	45	34
Malignant Oligodendroglioma	8	17	17	21	20	6		2	7	15	13	12	3
Oligodendroglial GMB	4	2	2	5	0	1		3	1	3	0	2	3
Smith 1983													
Indeterminate	72	58	69	60	81	59		64	46	54	61	52	40
Grade A	0	2	0	1	0	8		8	2	9	0	14	6
Grade B	8	24	13	10	5	8		3	15	21	19	20	23
Grade C	5	0	0	6	2	5		3	12	3	2	4	9
Grade D	12	3	2	16	4	6		8	9	7	5	5	6
Kros 1988													
Indeterminate	72	58	69	60	81	81		64	46	54	61	52	40
Oligodendroglioma	0	2	0	1	0	0		8	2	9	0	14	6
Malignant Oligodendroglioma	17	37	29	23	15	15		16	12	3	6	4	2
Oligodendroglial GBM	12	3	2	16	4	4		8	9	7	5	5	6
WHO 1993													
Indeterminate	82	67	45	70	86	86		60	64	44	65	43	45
Oligodendroglioma	1	28	33	2	6	6		19	7	31	25	48	36
Anaplastic Oligodendroglioma	28	25	19	21	9	9		34	23	39	33	19	35
WHO 1993 (after ignoring pleomorphism and anaplasia as grading criteria)													
Indeterminate	13	12	11	15	10	19		22	31	20	22	10	24
Oligodendroglioma	64	49	54	52	70	48		56	38	50	49	55	40
Anaplastic Oligodendroglioma	23	39	35	34	19	33		23	31	30	29	35	36

* One neuropathologist did not indicate the degree of prominent cellularity so his assessment could not be evaluated. The numbers indicate the percentage of patients classified into each grade after applying each grading scheme, separately, to the ratings endorsed by each pathologist.

limited number of findings, including high cellularity, mitotic activity, endothelial hypertrophy and/or proliferation, and necrosis, appear to be reproducible, especially when some preliminary explanation is provided. Based on this study, there also seem to be differences between surgical pathologists and neuropathologists in that the latter agree among themselves somewhat more frequently. This is particularly the case with respect to endothelial hypertrophy. This is not surprising and might be explained by the fact that, while features such as mitotic index and necrosis are key elements of grading in many tumor systems, microvascular changes remain in some way unique to brain tumors and neuropathology. Thus, while neuropathologists are critically trained to look for microvascular changes in CNS tissue, it would appear that some surgical pathologists are not.

We were able to obtain long follow-up in all but 1 of the patients included in the study and could therefore test the hypothesis if a relationship exists between histological features and survival. It is of note that overall survival in our patient sample (62% at 5 years, 38% at 10 years) is very similar to that previously reported in oligodendroglial tumors of all grades (5). As in previous studies, both univariate and multivariate analyses found age to be the strongest independent predictor of survival.

Among reproducible histological parameters, several appear to be statistically associated with cause-specific survival in univariate analyses. This was particularly true of mitotic activity, an observation previously reported by Burger et al (11) but at odds with several other studies (10, 13, 17, 18). Although we found no significant association between cause-specific survival and number of mitoses per 10 HPF based on consensus, this likely resulted from a large number of cases being classified as indeterminate regarding actual mitotic counts using our strict definition of consensus (Fig. 6). It is of note that

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mitotic index appears to be highly associated with cause-specific survival based on single neuropathologist assessment. Thus our data indirectly confirm what has been previously found in several studies, that a statistically significant association exists between survival and proliferative activity as measured by either MIB 1 labeling (19–22) or flow cytometric S-phase analysis (23).

Microvascular changes, including both endothelial hypertrophy and endothelial proliferation, appear to be highly associated with cause-survival in univariate analyses, while in multivariate analysis based on consensus among neuropathologists, only endothelial proliferation was independently associated with cause-specific survival after adjusting for patient age. Burger et al (11) as well as Schiffer et al (22) also found endothelial proliferation to be significantly associated with survival by univariate analysis, but not on multivariate analysis. The studies of Daumas-Duport (17, 18) showed that ''endothelial hyperplasia'' correlated significantly with survival, while glomeruloid vessels and/or an endothelial proliferation did not. The latter authors proposed a grading scheme based on the presence of ''endothelial hyperplasia'' and/or presence of contrast enhancement on imaging, which divides oligodendrogliomas into low- and high-grade forms. A direct comparison between our study and these studies is not possible since although ''endothelial hypertrophy'' and ''endothelial hyperplasia'' may closely correspond, no definition of endothelial proliferation is provided. Furthermore, given the span of years (1960 through 1990) from which we selected our cases, neuroimaging data were insufficient to address the issue of contrast enhancement and its effect upon survival. Lastly, as previously reported (11), necrosis was also significantly associated with survival, but only by univariate analysis. The low frequency of this finding (13 %) may in part explain its lack of association with survival in multivariable modeling.

Our results support the present stratification of oligodendroglial tumors by both the 1993 and 2000 WHO classifications (5), which divide them into low (grade II) and intermediate/high grade (grade III or ''anaplastic''). Unlike diffusely infiltrating astrocytic tumors, a distinction between oligodendrogliomas of grades III and IV does not appear to be possible solely on histologic grounds. Both mitotic index and microvascular changes appear to be equally associated with cause-specific survival in the hands of highly trained neuropathologists. A brisk mitotic index ($\geq$ 6 mitoses per 10 HPF) appears to be associated with significantly worse survival. Although this level of proliferative activity is not difficult to confirm, caution is necessary in the distinction of mitotic figures from pyknotic/apoptotic nuclei. It remains to be seen whether assessment of endothelial proliferation can be reproduced as reliably by general pathologists.

Our approach to classification and grading of oligodendrogliomas, based exclusively on histological grounds, provides important prognostic information relative to this tumor group. More of a challenge is the diagnosis and grading of mixed oligoastrocytomas, tumors which in our practice are every bit as frequent. The utility of molecular diagnostic methods in brain tumor classification and grading has attracted great interest (2, 24, 25). In addition to providing insights into tumorigenesis and progression, they may serve as an important adjunct if not a more accurate, objective and reproducible method of diagnosis. At present, however, histology remains the ''gold standard'' in every day practice.

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